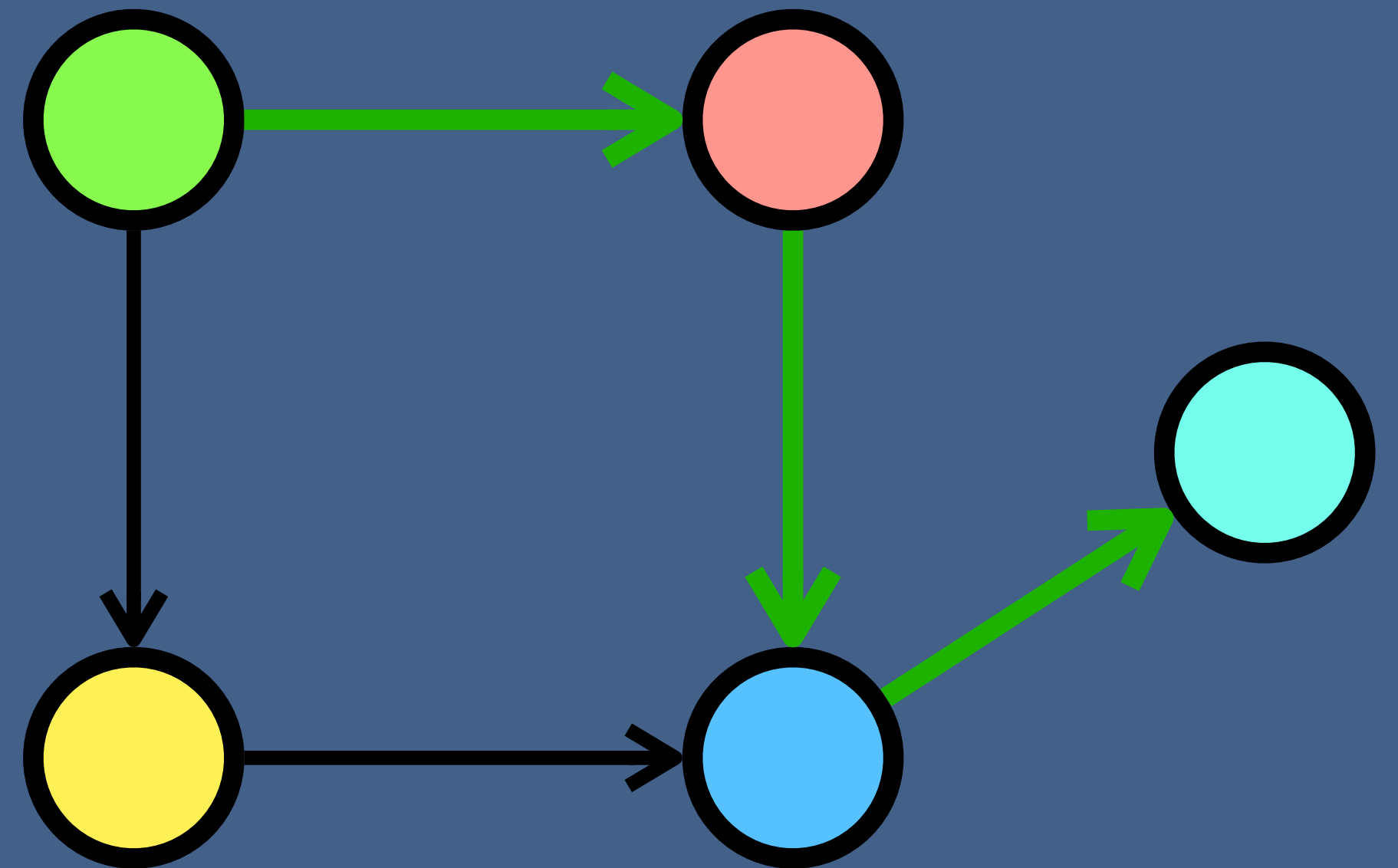


# Single Source Shortest Path Problem (SSSPP)

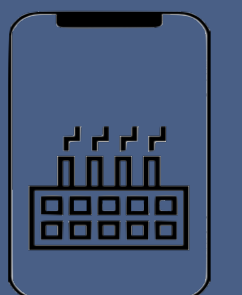
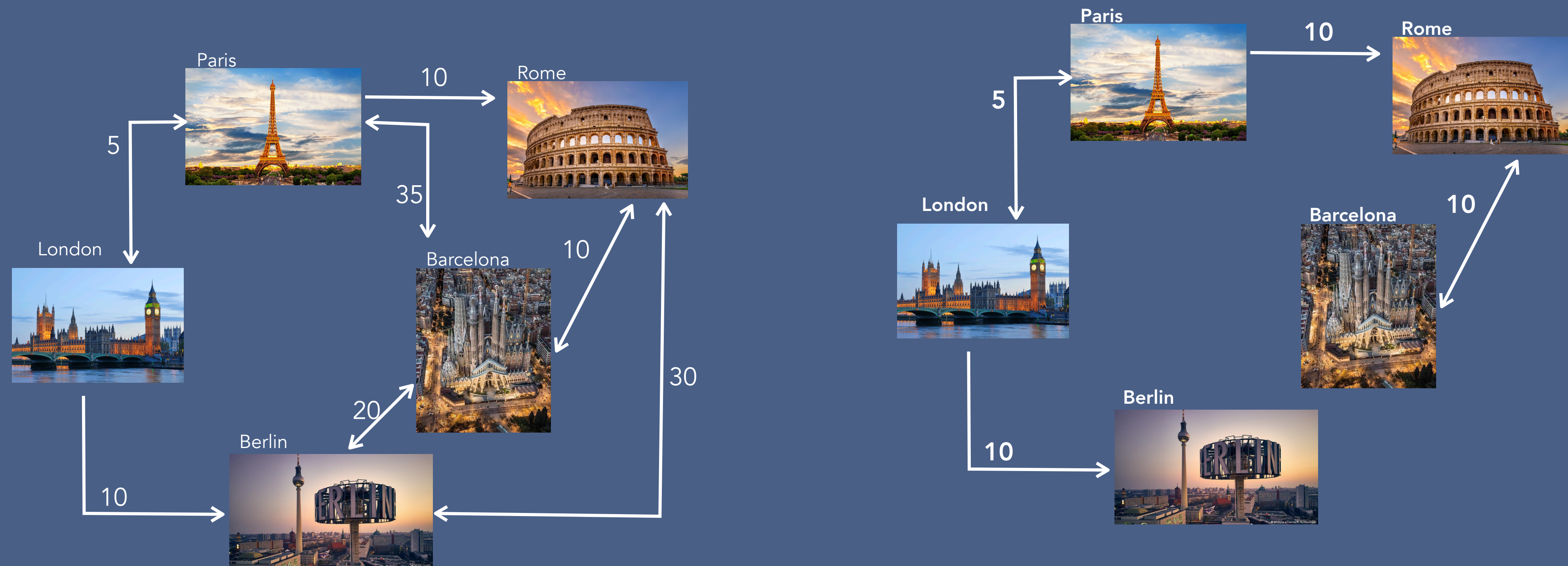


# Single Source Shortest Path Problem (SSSPP)

A single source problem is about finding a path between a given vertex (called source) to all other vertices in a graph such that the total distance between them (source and destination) is minimum.

The problem:

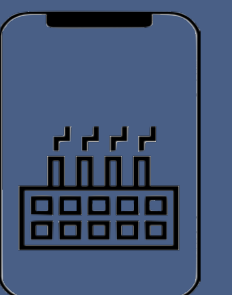
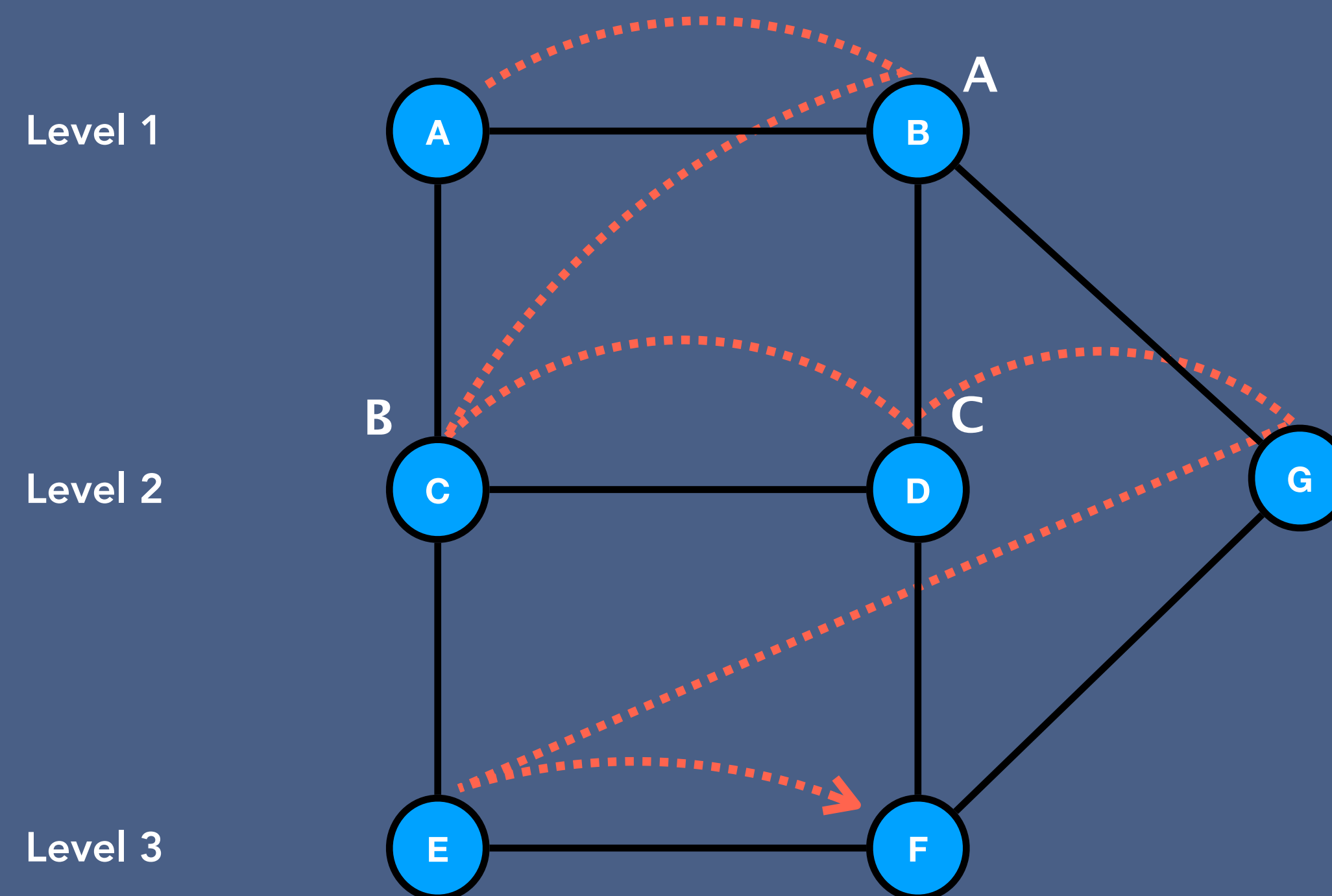
- Five offices in different cities.
- Travel costs between these cities are known.
- Find the cheapest way from head office to branches in different cities



# Single Source Shortest Path Problem (SSSPP)

- BFS
- Dijkstra's Algorithm
- Bellman Ford

## BFS - Breadth First Search





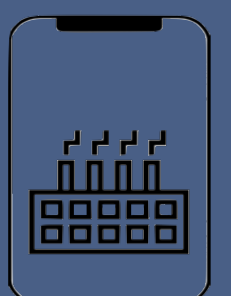
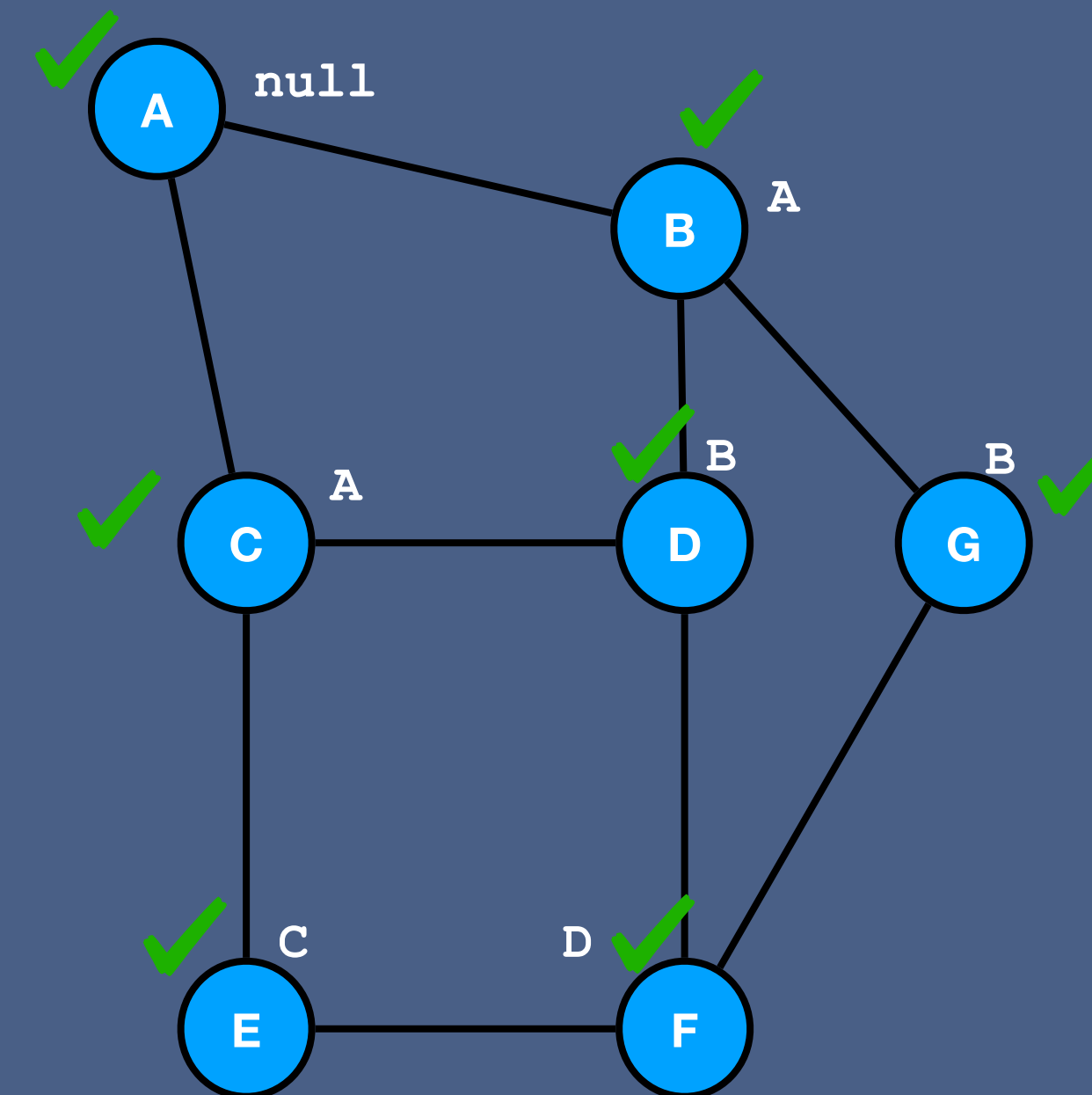
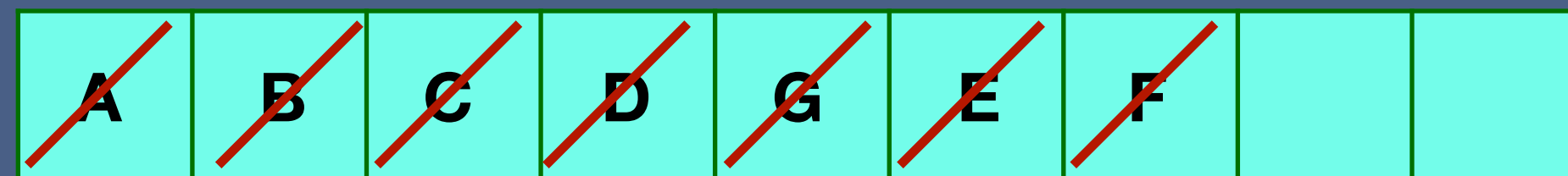
# BFS for SSSPP

BFS

```
enqueue any starting vertex
while queue is not empty
  p = dequeue()
  if p is unvisited
    mark it visited
    enqueue all adjacent unvisited vertices of p
    update parent of adjacent vertices to curVertex
```

A B C D G E F

Queue



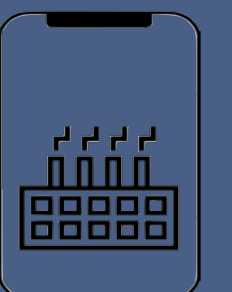


# Time and Space Complexity - BFS for SSSPP

```
create queue .....➔ O(1)
enqueue any starting vertex.....➔ O(1)
while queue is not empty .....➔ O(V)
  p = dequeue() .....➔ O(1)
  if p is unvisited .....➔ O(1)
    mark it visited.....➔ O(1)
    enqueue all adjacent unvisited vertices of p .....➔ O(adj)
    update parent of adjacent vertices to curVertex.....➔ O(1)
```

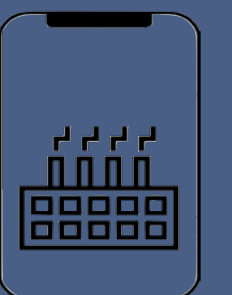
**Time Complexity :  $O(E)$**

**Space Complexity :  $O(V)$**

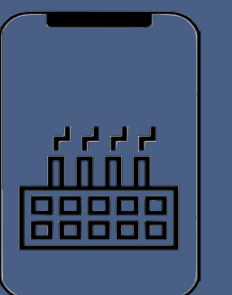
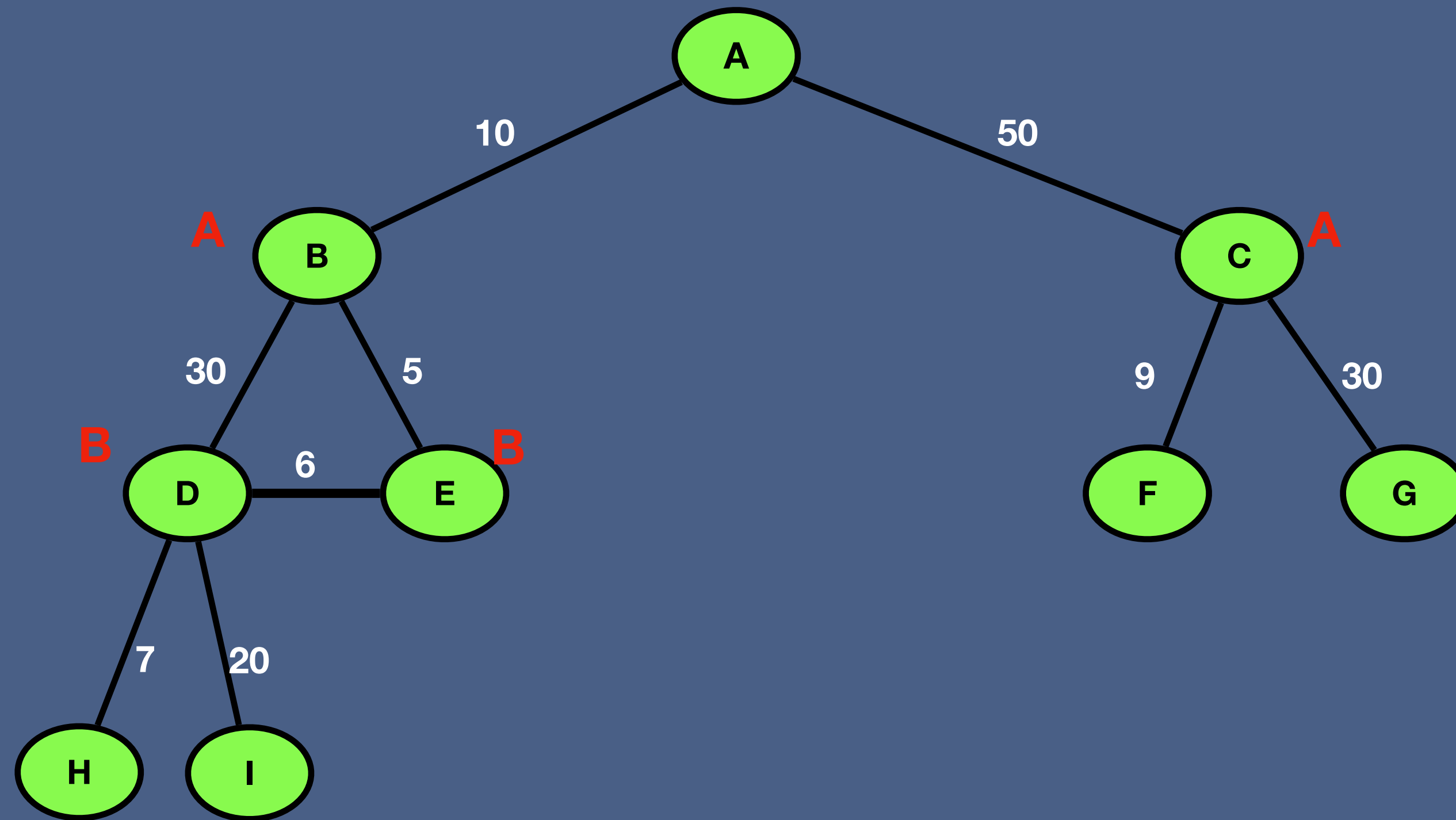


# Why BFS not work with weighted Graph?

| Graph Type                       | BFS |
|----------------------------------|-----|
| Unweighted – undirected          | OK  |
| Unweighted – directed            | OK  |
| Positive – weighted – undirected | X   |
| Positive – weighted – directed   | X   |
| Negative – weighted – undirected | X   |
| Negative – weighted – directed   | X   |



# Why BFS not work with weighted Graph?





# Why does DFS not work SSSPP?

DFS has the tendency to go "as far as possible" from source, hence it can never find "Shortest Path"

